1. **Differentiate** between Accuracy and Precision of a measuring instrument
2. The expected value of the voltage across a resistor is 80 V. However, the measuring gives a value of 79 V. **State** the percentage accuracy?

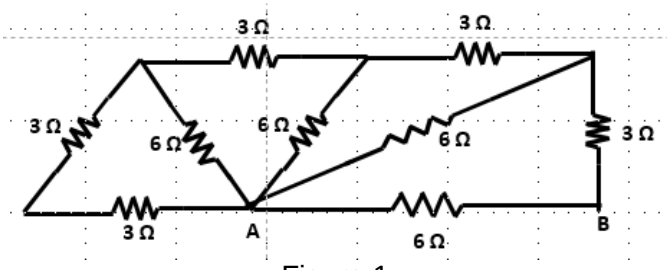


Figure 1

3. For the circuit in figure 1, **state** the resistance between terminals A and B
4. A pure inductor does not consume any real power. **Explain?**
5. The field coil of a shunt motor has a resistance of 45 Ω at 20 °C. **Give** the average temperature of the winding at the end of the run when the resistance is increased to 48.5 Ω. Temperature coefficient of resistance is 0.004/ °C at 0 °C
6. The flux produce in the air gap between two electromagnetic poles is 5×10^{-2} wb. If the cross sectional area of the air gap is $0.2\ m^2$. **Give** the value of the reluctance
7. **Why** is the star point of the supply transformer earthed?

8. A 230 V, 50 Hz voltage is applied to a coil of $L = 5$ H and $R = 2\ \Omega$ in series with a capacitance C. **What** value must C have in order that the voltage across the coil shall be 250 V

9. **State** the relationship between currents and the relationship between voltages in a delta connected system?

10. **What** is the necessity of earthing in domestic wiring?

11. **What** is the need of controlling and damping torques in analogue instruments

12. **What** is significance of root mean square (rms) value of alternating quantity?

13. **What** do you mean by a balanced 3-phase system?

14. **What** is power factor and its significance in domestic wiring system?

15. **Why** are domestics appliances are connected in parallel?

Section B

Answer **eight** (8) questions $8 \times 5 = 40$ **marks**

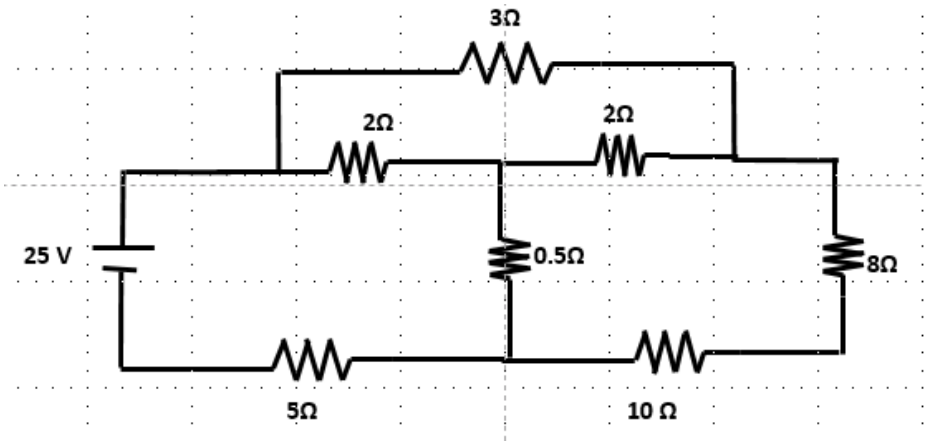


Figure 2

1. Using Kirchhoff's Law, **calculate** the power dissipated in the 8 Ω resistor in Figure 2

2. **Convert** all the voltage sources into its equivalent current source and **calculate** the current in the 0.25 Ω resistor in Figure 3 by **Norton's Theorem**

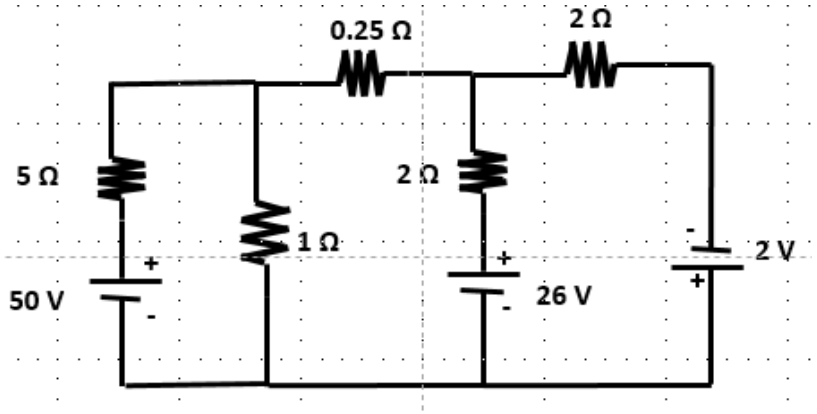


Figure 3

3. A network is arranged as shown in Figure 4. **Find** the voltage across the 8 Ω resistor by **Superposition theorem**.
4. Figure 5 depicts a circuit containing a voltage source and current source V_0 and I_0 respectively connected to two resistors R_1 and R_2. **Evaluate** I_1, I_2 and V_A. Given $V_0 = 10V$, $I_0 = 5A$, $R_1 = 10\Omega$, $R_2 = 5\Omega$ by **Nodal Analysis**

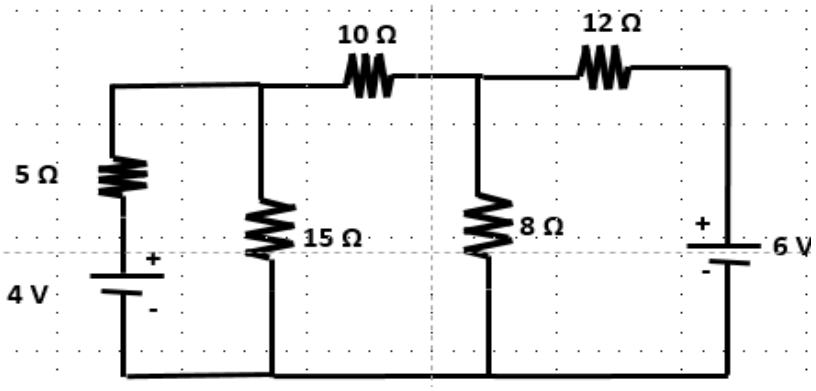


Figure 4

5. Using **Thevenin's theorem**, **calculate** the current through the 15 Ω in Figure 4
6. A symmetrical trapezoidal waveform is shown in Figure 5. **Show that** the Average and RMS values are $0.666V_m$ and $0.745V_m$, respectively.

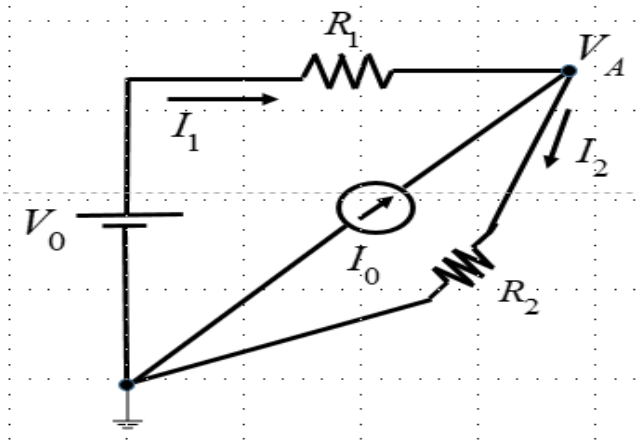


Figure 5

7. Two impedances $Z_1 = (10 - j15)\Omega$ and $Z_2 = (4 + j8)\Omega$ are connected in parallel. The supply voltage is 100 V, 25 Hz. **Calculate** the (i) the admittance, conductance and the susceptance of the combine circuit, and (ii) the total current drawn and its power factor.

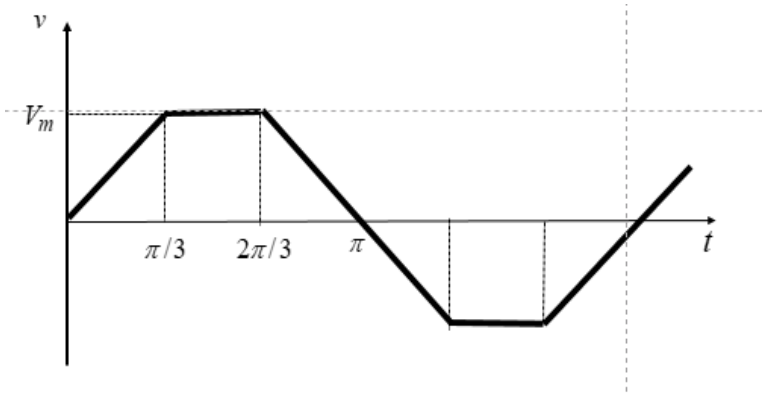


Figure 6

8. A coil having 200 turns and the area of cross section 250 square meters is rotated about its axis perpendicular to that of uniform magnetic field of strength 0.5 tesla at a speed of 1200 r.p.m. **Determine**

(a) Equation for the instantaneous induced emf

(b) Instantaneous value of induced emf in the following cases

(i) The plane of the coil is parallel to the field

(ii) The plane of the coil is perpendicular to the field

(ii) The plane of the coil is inclined at angle of 45° to the field

9. **Design** a simple circuit to permit an ammeter of resistance 0.5 Ω and full scale deflection of 15 A to measure 100A, 150A, and 200A.
10. A 3 phase, 400 V supply is given to a balanced star connected load of impedance $Z = (8 + j6)\Omega$ in each branch. **Calculate** the line current, power factor and total power. What will be the power factor regarding balanced delta connected load of the same impedance
11. **Explain** with the help of a diagram, **how** a dc generator works. **Why** is commutator used in it?
12. A 6 KVA, 3 phase induction motor has a power factor 0.7 lagging. **Calculate** the KVAr rating of a capacitor to improve the power factor to 0.89 lagging.